Department of Information Systems and Technologies

CTIS 365: Applied Data Analysis

Semester: SPRING 2025-2026

Lab Guide #7

OBJECTIVES: Probability (pnorm, dnorm), Correlation
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Q1. Find each of the following probabilities for a normal distribution.

a. $p(z > 1.05)$

b. $p(z > -0.40)$

c. $p(z < 0.40)$

d. $p(z < -1.20)$

Q2. For a normal distribution with a mean of $\mu = 40$ and a standard deviation of $\sigma = 9$, find the proportion of the population corresponding to each of the following.

a. Scores greater than 55.

b. Scores less than 35.

c. Scores between 50 and 70.

Q3. Sat verbal scores are normally distributed with **mean** of 420 and a **standard deviation** of 65.

a. What percentage of students score less than 350?

b. What percentage of students score between 350 and 500?

Q4. Calculate the **Spearmen correlation** for the following set of scores.

```
x <- c(0,3,2,5,0)
y <- c(1,3,3,2,1)
```

Q5. Calculate the **Pearson correlation** by converting the following sample data to z-score.

```
x <- c(2,5,1,3,4)
y <- c(3,1,1,2,5)
```